

Literature status for arXiv:1805.10203: Gaussian multi-bubble conjecture

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Status

This is a lightweight literature-status packet. The extracted target from Steven Heilman's *The Structure of Gaussian Minimal Bubbles* [1] is already answered by separate work of Milman and Neeman [2, 3]. It is not a new proof or partial result from this run.

Source Question

The source paper states the Gaussian Multi-Bubble Problem as Problem 1.2 and the Gaussian Multi-Bubble Conjecture as Conjecture 1.3. In source notation, for $m \geq 3$ and prescribed positive Gaussian volumes $a_1, \dots, a_m$ summing to 1, Problem 1.2 asks for a measurable partition

$$\Omega_1, \dots, \Omega_m \subset \mathbb{R}^{n+1}, \quad \gamma_{n+1}(\Omega_i) = a_i,$$

minimizing total Gaussian interface area

$$\sum_{1 \leq i < j \leq m} \int_{\partial\Omega_i \cap \partial\Omega_j} \gamma_n(x) dx.$$

Conjecture 1.3 predicts that every minimizer is, up to translation, the Voronoi partition determined by the vertices of a centered regular simplex:

$$\Omega_i = y + \{x \in \mathbb{R}^{n+1} : \langle x, z_i \rangle = \max_{1 \leq j \leq m} \langle x, z_j \rangle\}.$$

Supporting Answer

Milman and Neeman's supporting paper proves the Gaussian Multi-Bubble Theorem. In the arXiv version, Theorem 1.1 states that the Gaussian Multi-Bubble Conjecture holds for all $2 \leq q \leq n+1$, and Theorem 1.2 states uniqueness up to null sets of the corresponding simplicial q -clusters for prescribed positive Gaussian measures.

This directly answers the source's Conjecture 1.3 in the standard dimension range where $m = q \leq n+1$ regular simplex clusters exist. The source also explicitly records this provenance: after Heilman's first arXiv upload, Milman and Neeman uploaded arXiv:1805.10961, solving Conjecture 1.3 unconditionally.

Search And Evidence

Cheap run-index search for arXiv:1805.10203, Gaussian minimal bubbles, Gaussian Multi-Bubble Problem, Gaussian Multi-Bubble Conjecture, simplicial cluster, and plurality-is-stablest found no existing packet. Web/literature search found the Annals of Mathematics publication page for Milman–Neeman, which states that the paper establishes the Gaussian Multi-Bubble Conjecture and gives the DOI 10.4007/annals.2022.195.1.2. The local source bibliography cites arXiv:1805.10961 as *The Gaussian multi-bubble conjecture*.

Scope Limitation

This packet only records the literature status of the Gaussian multi-bubble problem extracted from arXiv:1805.10203. It does not address Euclidean, spherical, hyperbolic, anisotropic, or stability variants of multi-bubble isoperimetry unless those variants are explicitly covered by the cited Milman–Neeman theorem.

References

- [1] Steven Heilman, *The Structure of Gaussian Minimal Bubbles*, arXiv:1805.10203, 2018.
- [2] Emanuel Milman and Joe Neeman, *The Gaussian Multi-Bubble Conjecture*, arXiv:1805.10961, 2018.
- [3] Emanuel Milman and Joe Neeman, *The Gaussian Double-Bubble and Multi-Bubble Conjectures*, Annals of Mathematics 195 (2022), no. 1, 89–206. DOI: 10.4007/annals.2022.195.1.2.